Basic Principles of Medicine 2
Module : Digestion and Metabolism
DLA

Autonomics of the Abdomen

Nervous System Integration

Somatic

SOMATIC SENSORY

Receptors in **BODY WALL** for **conscious** sensation:

- Pain
- Pressure
- Touch
- Temperature
- Proprioception

Somatic Sensory Afferents →

CNS

Somatic Motor Efferents →

SOMATIC MOTOR

- **Voluntary** control of **Skm**

PNS   PNS

Visceral Afferents

Receptors in ORGANS and **TISSUES**:

- Distension
- Inflammation
- Ischemia
- Physiological Changes
- Abnormal Muscle Spasm

Visceral Sensory Afferents →

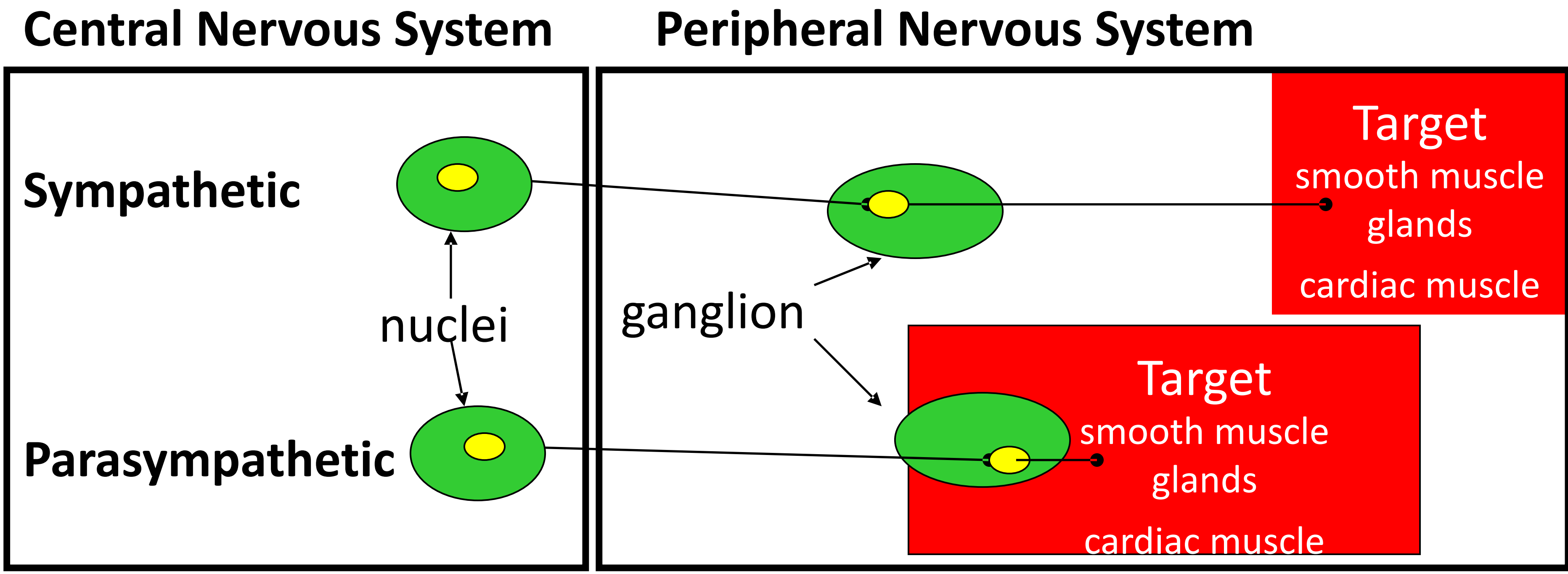
Visceral Motor Efferents →

Visceral Efferents

Involuntary control of

- SM
- Cardiac muscle
- Glands

Divisions of the Autonomic Nervous System



Preganglionic =
Myelinated

Postganglionic =
UNmyelinated

PREganglionic

Parasympathetic Outflow

Brainstem

CN **3**,**7**,**9**,**10**

Sympathetic

Outflow

Lateral horn of

Thoracolumbar

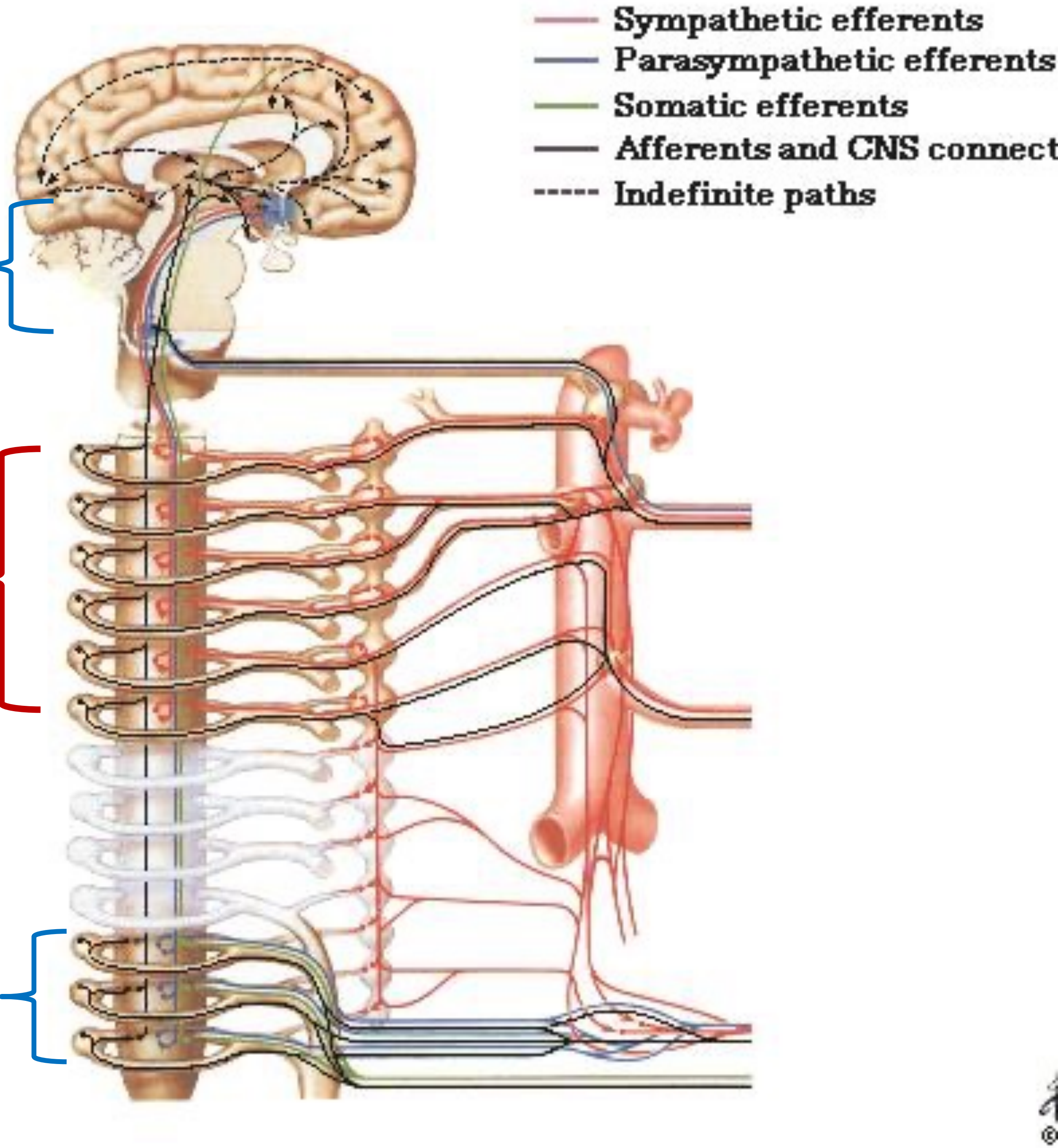
spinal cord

T1 –L2

Sacral spinal

segments

S2-S4



POSTganglionic

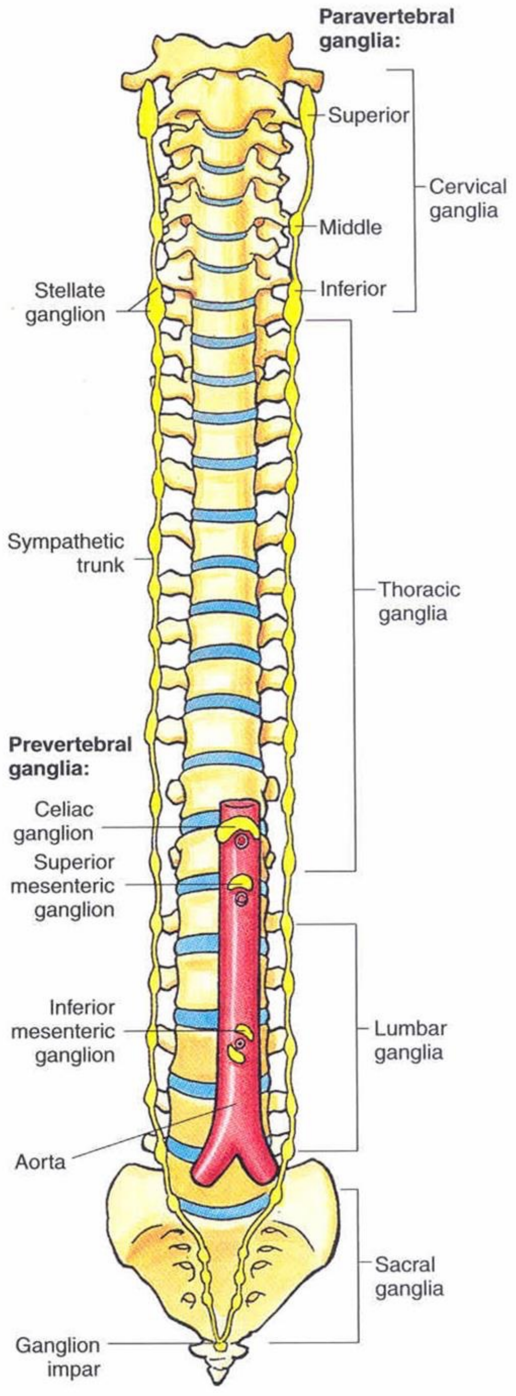
Paravertebral / Sympathetic Chain Ganglia

- **3** in **Cervical** region
- **One** per **Ventral ramus**

Preaortic / Prevertebral Ganglia

@ roots of major arteries branching from Abdominal Aorta

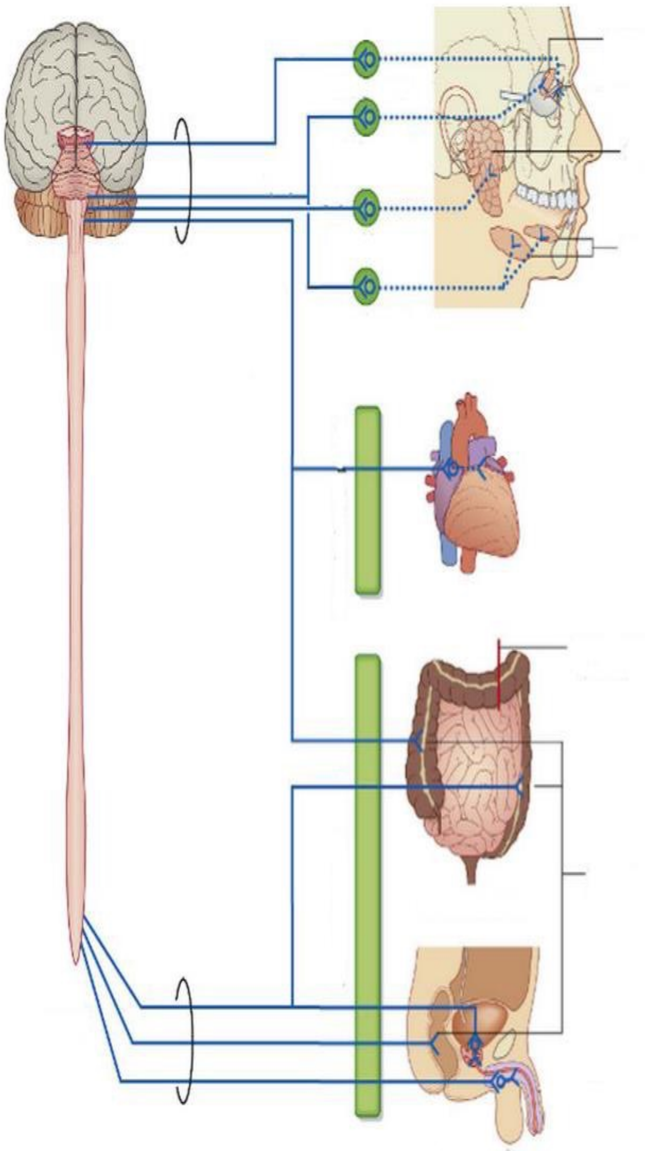
- **Celiac**
- **Superior Mesenteric**
- **Inferior Mesenteric**
- **Renal**



Postganglionic Parasympathetic are found **scattered near or in** the **walls** of **target** organs

- **Discrete Parasympathetic ganglia** are only **found in** the **head** associated **with** parasympathetic **CNs**

- **Parasympathetic ganglia in** the **GI synapse on** the **neurons of** the **Meissner submucosal** plexus **and Myenteric Auerbach** plexus, **both** constitute **Enteric Nervous System**



Functions of the ANS in the GI

Sympathetic NS

- **Vasoconstriction**
- **Constriction** of **Sphincters** for **Closure**

Parasympathetic NS

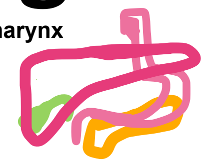
- **Increase** GI **motility**
- **Increase** GI **secretion**

Enteric Nervous System

- **Intrinsic** neuronal **circuit** consisting of **motor** and **sensory** neurons **organized into 2 interconnected plexuses** (**Myenteric** and **Meissner's**)
- The **Enteric Nervous** System is generally **independent**, **but receives input from both** the **Sympathetic** and **Parasympathetic** systems

Embryological Origins

Pharynx

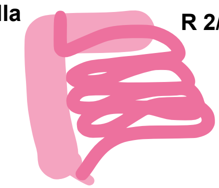


Proximal Major Duodenal Papilla

Derivatives of the Foregut

- **Pharynx** and its derivatives, **Lower respiratory** tract, **Esophagus**, **Stomach**, **Duodenum** up **to** the **major duodenal papilla** , **Liver**, **Biliary apparatus** and **Pancreas**

Distal Duodenal Papilla



R 2/3 of Transverse Colon

Derivatives of the Midgut

- **Duodenum distal to** the **major duodenal papilla**, **Jejunum** & **Ileum**; **Cecum** & **Appendix**, **Ascending** colon & **right 2/3** of **transverse** colon

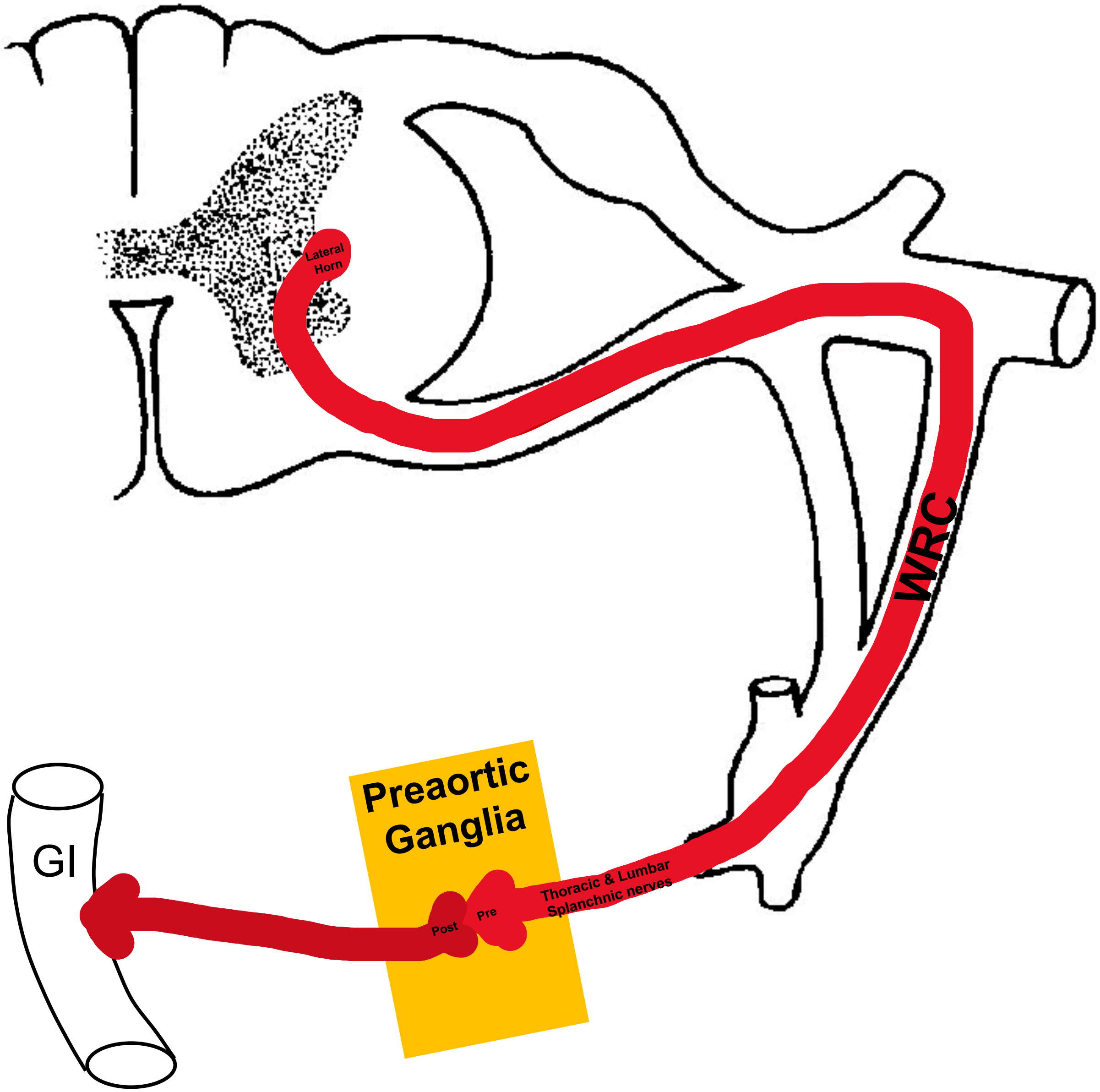
L 1/3 of Transverse Colon

Pectinate line

Derivatives of the Hindgut

- **Left 1/3** of **Transverse** colon, **Descending** colon, **Sigmoid** colon & **Rectum**, **Anal canal** up **to** the **pectinate line,** Urinary **bladder** & **most** of **Urethra**

Sympathetics to Gastrointestinal System



- **Preganglionic** fiber **enters** chain via **WRC**
- **Passes** on through **ganglion WITHOUT synapsing**
- **Still PREganglionic** fiber then **leaves** the **medial** aspect of the **ganglion** and travels **to** a **sympathetic ganglion on** the **Abdominal aorta- PREAORTIC** GANGLION
- **Postganglionic** fibers then **travel along plexuses on** the **arteries to** the **GI** tract

Sympathetic Splanchnics (Preganglionic) to GI

- **Greater Splanchnic (T5-T9) synapses in** the **Celiac** ganglion.
- **Lesser Splanchnic (T10-11) synapses in** the **Aortico-Renal** ganglion.
- **Least Splanchnic (T12) synapses in** the **Renal** plexus.
- **Lumbar Splanchnic (L1-L2) synapses in** the **Intermesenteric +Superior Hypogastric** plexus

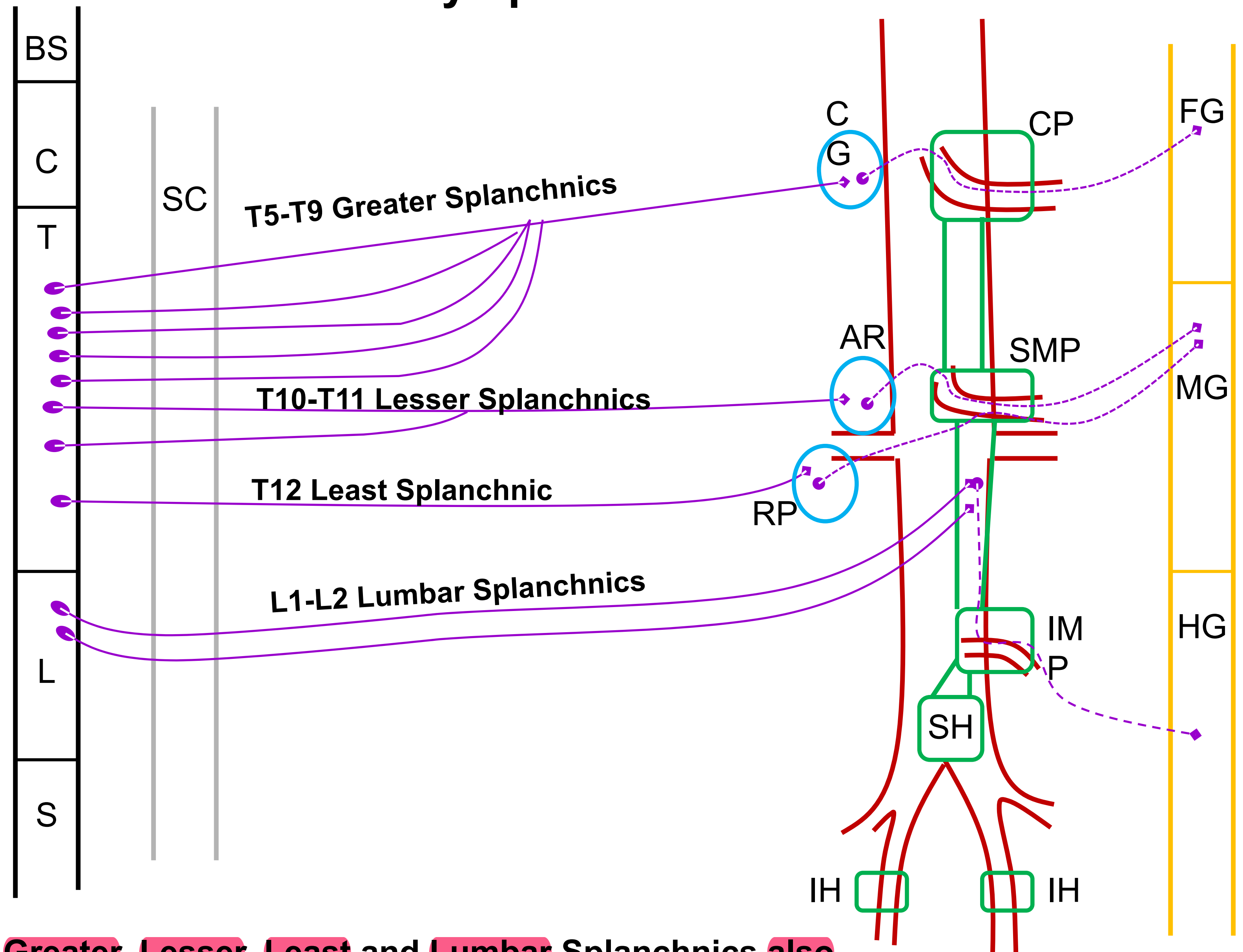
Sympathetic Postganglionic to the GI

Postganglionic **sympathetic** fibers **travel** on **Peri-arterial plexuses to target**

<u>**Arterial supply** is **based on embryonic origin**</u>

- **Foregut = Celiac** artery
- **Midgut**- =**Superior Mesenteric** artery
- **Hindgut = Inferior Mesenteric** artery

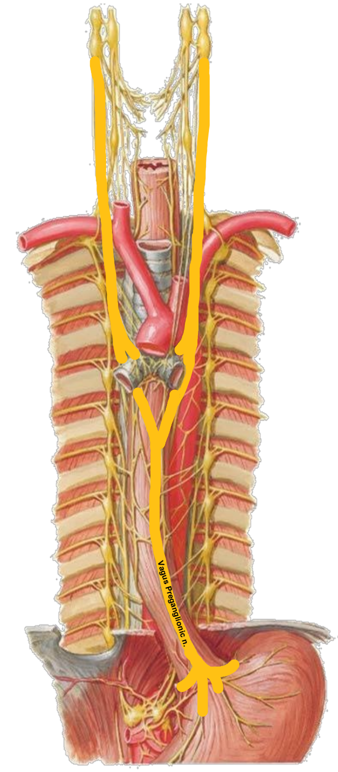
Sympathetics



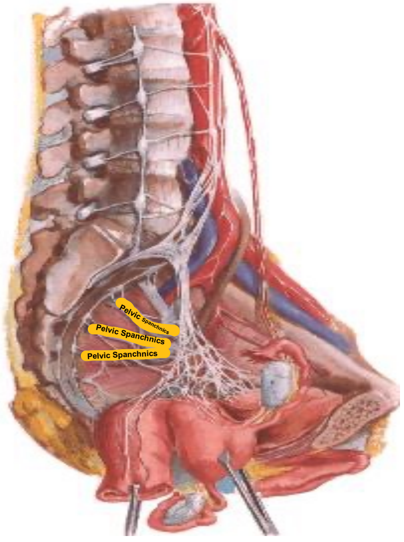
Note that Greater, Lesser, Least and Lumbar Splanchnics also provide PREganglionics to Kidneys, Ureters and Pelvic organs

Parasympathetics to GI System

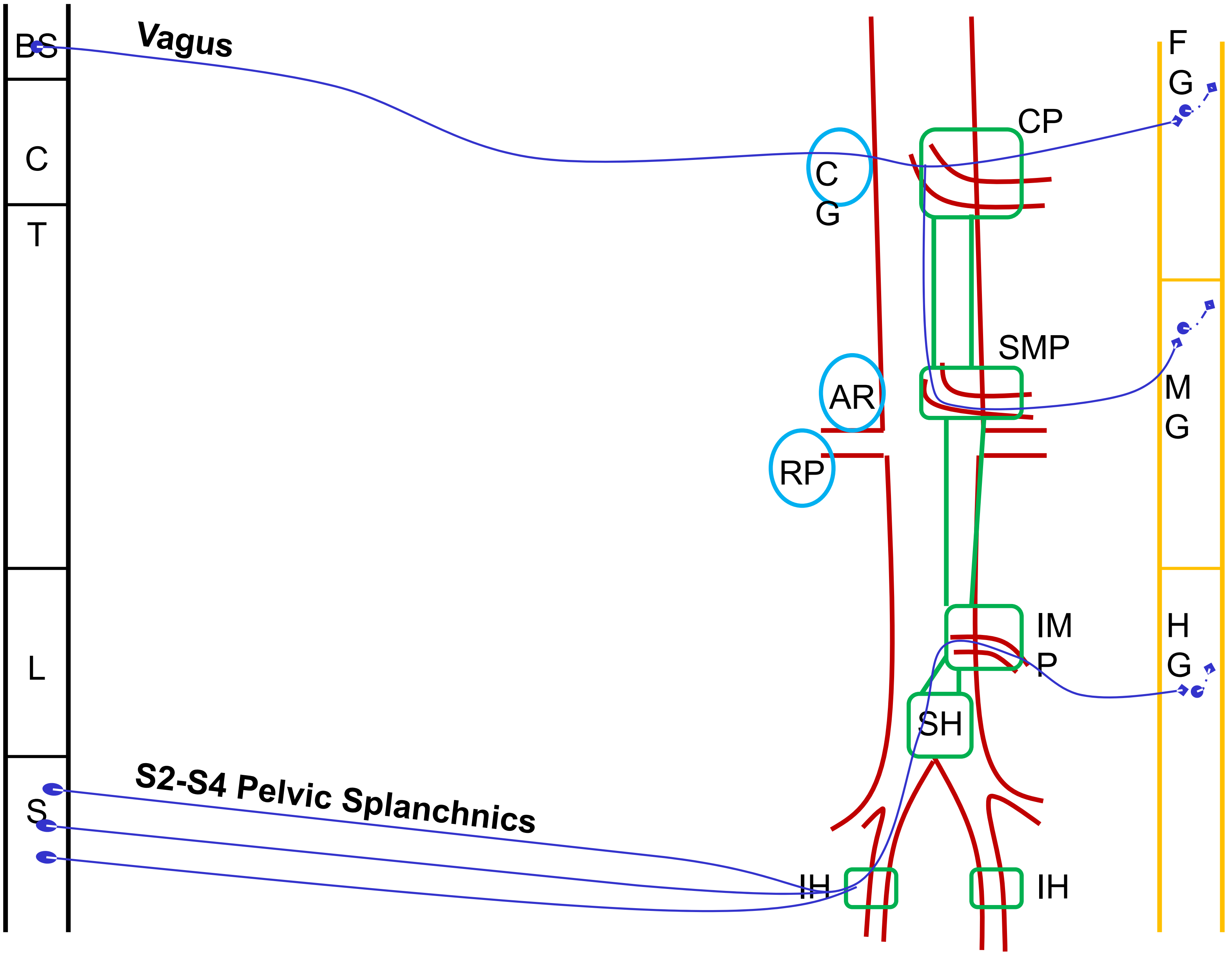
Vagus nerve = **Foregut** and **Midgut**



Pelvic Splanchnic S2-S4 = **Hindgut**



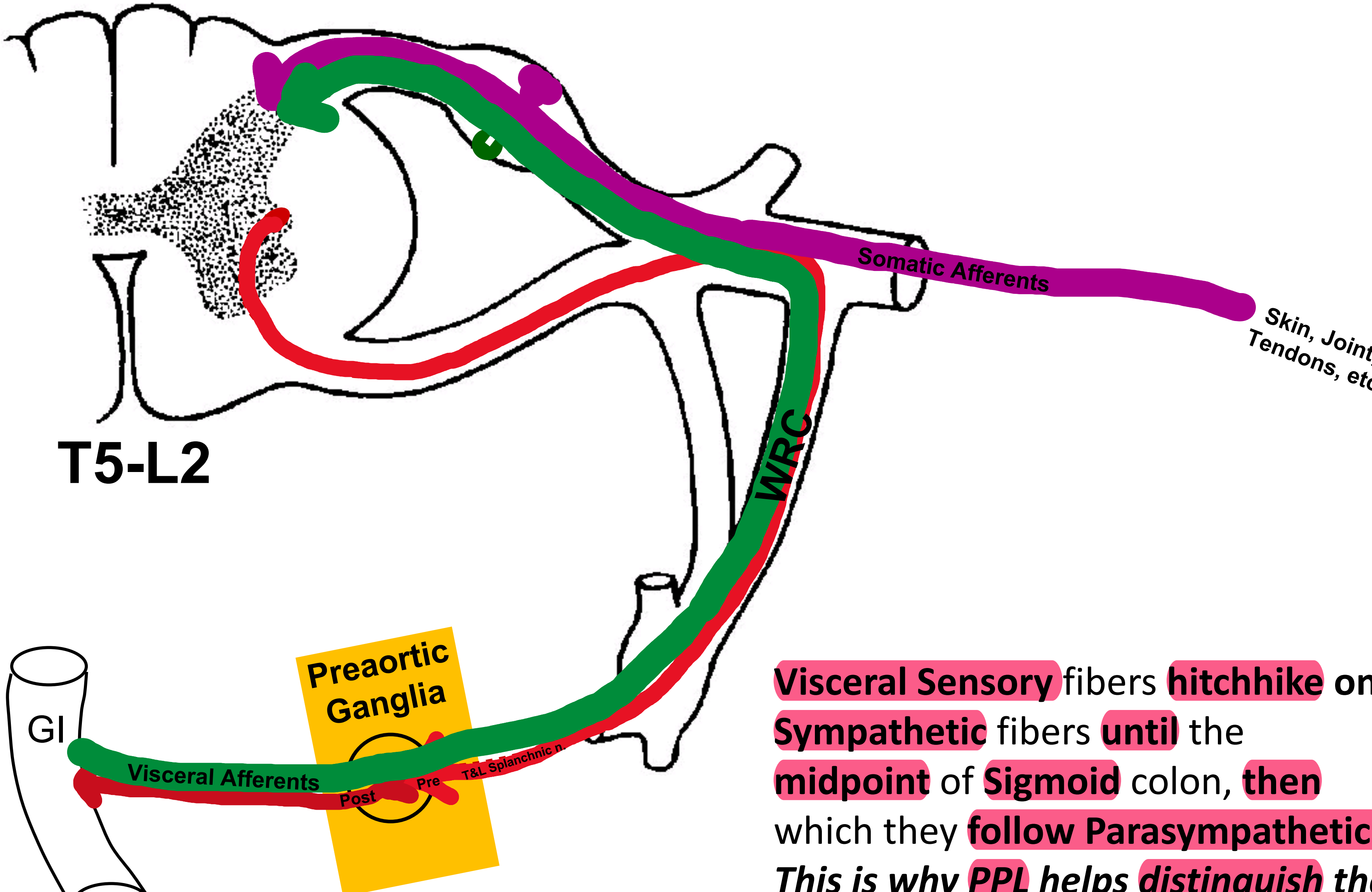
Parasympathetics



VISCERAL SENSORY FIBERS

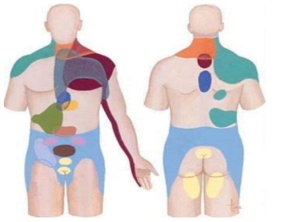
- **Visceral Afferents** fibers that **mediate pain** sensation (**e.g.** respond to **ischemia**, **inflammation** and **distention**) **travel with Sympathetic** fibers **until** the **midpoint of** the **sigmoid** colon where they **then follow Parasympathetics**

- **THEREFORE**, **by knowing** the **PREganglionic Sympathetic supply to** a particular **organ**, you **can** then be able to **determine where** in the body **pain** from those organs **will refer to**



Visceral Sensory fibers **hitchhike on Sympathetic** fibers **until** the **midpoint** of **Sigmoid** colon, **then** which they **follow Parasympathetics**. ***This is why PPL helps distinguish the difference***

Referred Pain



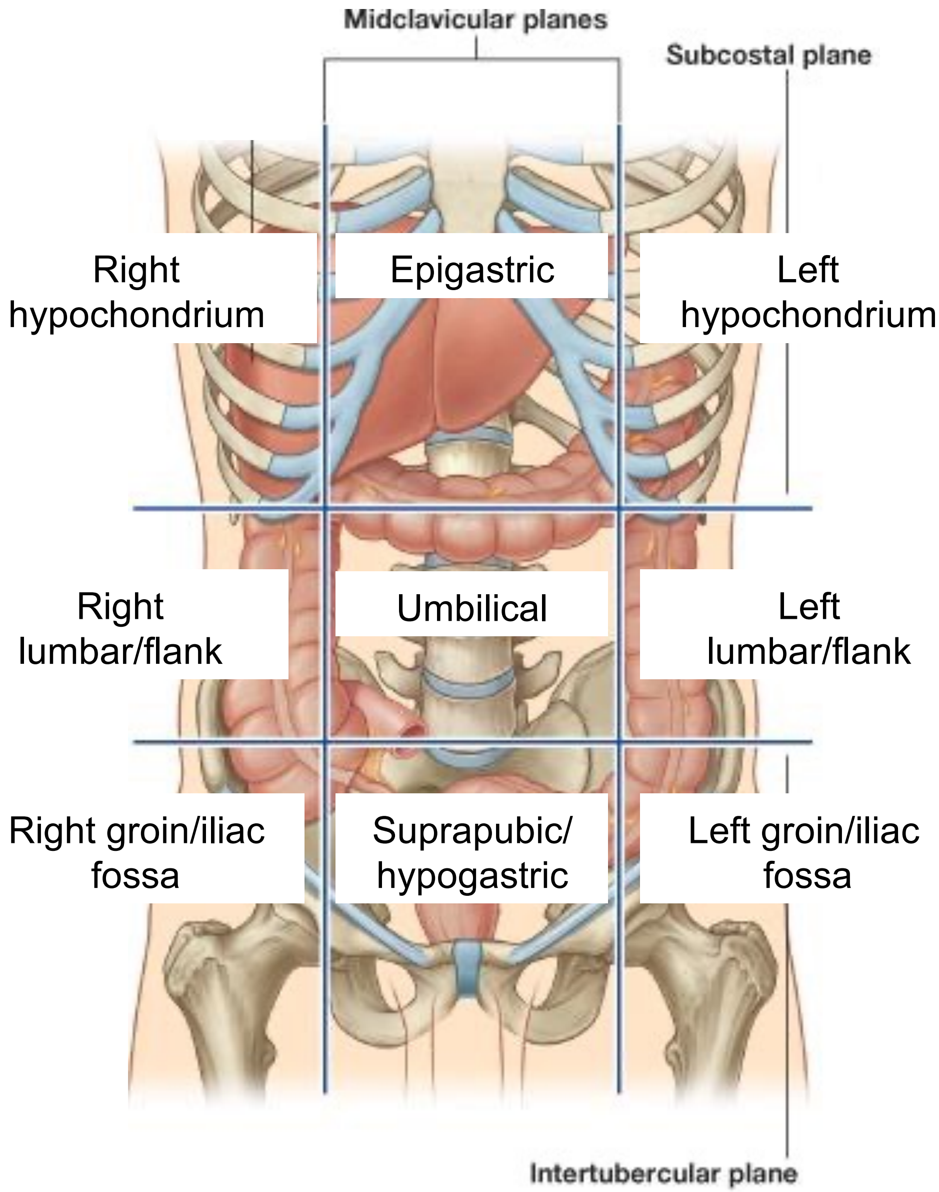
- **Sensory** information **enters** the **spinal cord** from one location, but is **interpreted by** the **CNS as coming from another location**

- For example, the **gut** (**visceral**) **have low sensory output** and **these** sensory **afferents converge @** the **same spinal cord** level **that receive sensory** afferents **from high output areas such as** the **body wall (somatic)**

- **As a result, the low sensory output** is **interpreted as coming from** the **area of high** sensory **output**

- **Pain can** also **be referred from one somatic region to another**.

Review


Midclavicular planes
Subcostal plane
Right hypochondrium
Epigastric
Left hypochondrium
Right lumbar/flank
Umbilical
Left lumbar/flank
Right groin/iliac fossa
Suprapubic/ hypogastric
Left groin/iliac fossa
Intertubercular plane

Referred Pain in the GI

- **Visceral Afferents follow** the **Autonomic** supply "backwards" **to** the **spinal cord**

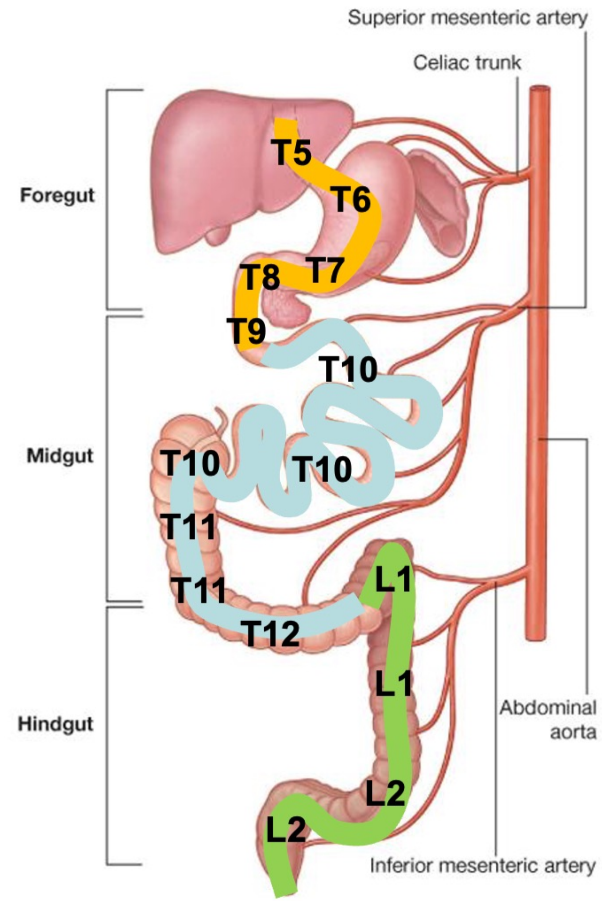
Generally, referred pain from

- **Foregut Structures =** pain here is **perceived as Epigastric/UQ pain** (**Sympathetics T5-9**)

- **Midgut Structures =** pain here is **perceived as Periumbilical pain** (**Sympathetics T10-12**)

- **Hindgut Structures ABOVE Sigmoid midpoint** pain here is **perceived as Flank/Groin/Lower Limb** (**Sympathetics L1-2**)

- **Hindgut Structures BELOW Sigmoid midpoint** perceived as **Perineal pain** (**parasympathetics S2-S4**)



GUT	DERIVATIVES	ARTERY	Preganglionic NERVE SUPPLY	
			Sympathetic	Parasympathetic
Foregut	ESOPHAGUS STOMACH PANCREAS LIVER GALL BLADDER BILE DUCT DUODENUM (cranial half)	CELIAC	GREATER THORACIC SPLANCHNIC NERVES	VAGUS
Midgut	DUODENUM (caudal half) JEJUNUM ILEUM CECUM & APPENDIX ASCENDING COLON TRANSVERSE COLON (proximal 2/3)	SMA	LESSER and LEAST THORACIC SPLANCHNIC NERVES	
Hindgut	TRANSVERSE COLON (distal 1/3) DESCENDING COLON SIGMOID COLON RECTUM ANAL CANAL(cranial part)	IMA	LUMBAR SPLANCHNIC NERVES	PELVIC SPLANCHNIC NERVES